

WELL COMPLETION REPORT: WELL 15/9-F-11B

OPERATOR: STATOIL ASA | LICENCE: PL 046 | FIELD: VOLVE

Well Name / UWI	15/9-F-11B	Country	Norway
Operator	Statoil ASA	Field Name	Volve
Spud Date	2007-08-24	Completion Date	2007-10-15
Surface Latitude	58.4394 N	Surface Longitude	1.8875 E
Total Depth (MD)	3200.0 m	Water Depth	86.0 m

1. Executive Summary

Well 15/9-F-11B was drilled as a production well in the southern part of the Volve Field. The objective was to penetrate the Jurassic Hugin Formation sandstone reservoir. Total depth of 3200.0 m MD was reached successfully in the Skagerrak Formation.

2. Geological & Stratigraphic Summary

The well penetrated standard North Sea stratigraphy including Nordland, Hordaland, Rogaland, Chalk Group, and Viking Group before entering the target Hugin Formation at 2420 m MD. The Skagerrak Formation was penetrated at 2750 m MD to total depth at 3200 m MD.

Formation	Top MD (m)	Base MD (m)	Lithology
Hugin Formation	2420.0	2750.0	Sandstone reservoir, fine to medium
Skagerrak Formation	2750.0	3200.0	Interbedded sandstone and shale

3. Operational Drilling Incidents & Mud Program

Mud Program: The 12-1/4 inch section from 2400.0 m to 2600.0 m MD was drilled using OBM with mud weight 1.45 SG. On 2007-09-12, while drilling through the Hugin Formation at 2450.0 m MD, partial mud losses of 15 bbl/hr were encountered. Pit level dropped steadily over 2 hours. Mitigation: Pumped LCM pill (50 bbl, 40 ppb LCM blend); losses reduced to seepage rate before drilling resumed.

4. Casing & Cementing Operations

Surface casing 13-3/8 inch (68 ppf) was set at 1200 m MD. Intermediate casing 9-5/8 inch (47 ppf) was set at 2600.0 m MD and cemented without issues with 280 bbl Class G + silica cement, achieving top of cement at 1800 m MD.

Casing Type	Shoe MD (m)	Size (in)	Weight (ppf)	TOC (m)
Surface	1200.0	13.375	68.0	Surface
Intermediate	2600.0	9.625	47.0	1800.0

5. Deep Section Operations & Lessons Learned

On 2007-09-20, at measured depth 2810.0 m MD in the Skagerrak Formation, the drillstring became mechanically stuck while tripping out through a tight hole section. Overpull of 40 klbs was observed and the crew was unable to rotate string. Action taken: Worked pipe free with jarring after 6 hours; wiper trip run before continuing.